

Karthik Aravind M

[LinkedIn](#) | [GitHub](#) | [GeeksForGeeks](#) | [LeetCode](#)
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EDUCATION

Kumaraguru College of Technology
B.E. Computer Science & Engineering
CGPA: 8.7 / 10

Class 12 (CBSE) – 96%
Class 10 (CBSE) – 94%

SKILLS

Languages

Java • Python • JavaScript • SQL

Frontend

React • HTML5 • CSS • Tailwind CSS

Backend

FastAPI • Node.js • REST APIs • WebSockets

Databases

MySQL • ChromaDB

AI / ML

LLMs • RAG • LangChain
LangGraph • NLP • ML

Developer Tools

Docker • Git • GitHub

INTERNSHIPS

AICTE-Eduskills |

Java Full Stack Development (Virtual)

June 2026 – August 2026

CERTIFICATIONS

Python for AI, ML & DL - [IIT Kanpur](#)
Full-Stack Web Dev Bootcamp – [Udemy](#)
Java Programming for Beginners – [Udemy](#)
RAG & MCP Fundamentals – [KodeKloud](#)

ACHIEVEMENTS

Mahatma Gandhi Merit Scholarship - [2024-25](#)
Mahatma Gandhi Merit Scholarship - [2025-26](#)

SUMMARY

Pre-final year Computer Science student with expertise in Data Structures & Algorithms, Full-Stack Development, Backend Engineering, and Generative AI. Experienced in designing RAG pipelines, real-time AI systems, scalable REST APIs, and containerized applications using React, FastAPI, Docker, and modern LLM frameworks.

PERSONAL PROJECTS

VoicelQ [Live Demo](#)

- Built a real-time AI voice customer support agent using **FastAPI** WebSockets, **Faster-Whisper**, **Piper TTS**, and **LangGraph** with multilingual speech-to-speech conversations and conversational memory.
- Implemented RAG-based knowledge retrieval, barge-in interruption, sentiment-driven human escalation, and a React frontend with live transcripts and session summaries.

Multimodal AI Study Assistant [Live Demo](#)

- Developed a full-stack AI study platform supporting PDF, image, audio, and video understanding using **OCR**, **Whisper**, **Vision LLMs**, and **RAG** with **ChromaDB**.
- Built AI-powered quiz generation, personalized notes, learning analytics, weak-topic detection, **JWT authentication**, and an admin dashboard.

DocMind [Live Demo](#)

- Developed a secure **RAG** platform for PDF question answering using **ChromaDB** vector search and **Groq LLMs** with context-grounded responses.
- Implemented **JWT authentication**, **RBAC**, **AI guardrails** (prompt injection detection and PII redaction), analytics dashboard, and **Docker**-based deployment.

AI Email Classification System [Live Demo](#)

- Built a full-stack email classification platform combining **Regex**, **Naive Bayes**, and **LLM-based** zero-shot classification through a majority-voting ensemble.
- Developed a scalable **React-FastAPI** architecture with **ChromaDB** and **Groq APIs** for low-latency semantic email classification.

Multi-Agent DevAssistant [Live Demo](#)

- Built a **Multi-Agent Developer Assistant** using **LangGraph** and **Groq LLaMA 3.3-70B**, where six specialized AI agents (Planner, Architect, Coder, Reviewer) autonomously generate production-ready codebases with a self-healing retry loop.
- Developed a **React-FastAPI** pipeline with **per-file LLM code generation**, **ChromaDB** codebase RAG, and real-time **SSE streaming** to handle token limits and deliver projects end-to-end.